

Module 13: Caching & Performance

Duration: 3-4 days | **Level:** Advanced

Cache Drivers

```
CACHE_STORE=database    # Learning
# CACHE_STORE=redis     # Production
```

```
php artisan cache:table && php artisan migrate
```

Basic Cache Operations

```
Cache::put('key', 'value', now()->addMinutes(10));
$value = Cache::get('key', 'default');
Cache::forget('key');
Cache::flush();
```

Remember Pattern (TaskForge DashboardService)

```
return Cache::remember(
    "dashboard.stats.{ $user->id }",
    now()->addMinutes(5),
    fn () => $this->computeStats($user),
);
```

Invalidate when data changes:

```
Cache::forget("dashboard.stats.{ $user->id }");
```

Query Optimization

1. **Eager loading:** with(['project', 'assignees'])
 2. **Select only needed columns:** select(['id', 'title'])
 3. **Chunk large datasets:** Task::chunk(100, fn (\$tasks) => ...)
 4. **Database indexes** on frequently queried columns
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Laravel Telescope (Debugging)

```
composer require laravel/telescope --dev
php artisan telescope:install
```

Monitors queries, requests, jobs, cache, mail.

Production Performance Checklist

- [] php artisan config:cache
- [] php artisan route:cache
- [] php artisan view:cache
- [] php artisan event:cache

● []composer install --optimize-autoloader --no-dev

- [] Redis for cache, sessions, queues

● [] OPcache enabled on PHP

Exercises

1. Cache project list for 10 minutes per user.
 2. Use `Cache::tags()` (Redis only) for grouped invalidation.
 3. Profile a page with Telescope and fix one N+1 query.
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Next Module

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